



Fig. 4: Author's IR sensor transmitter fixed on chair

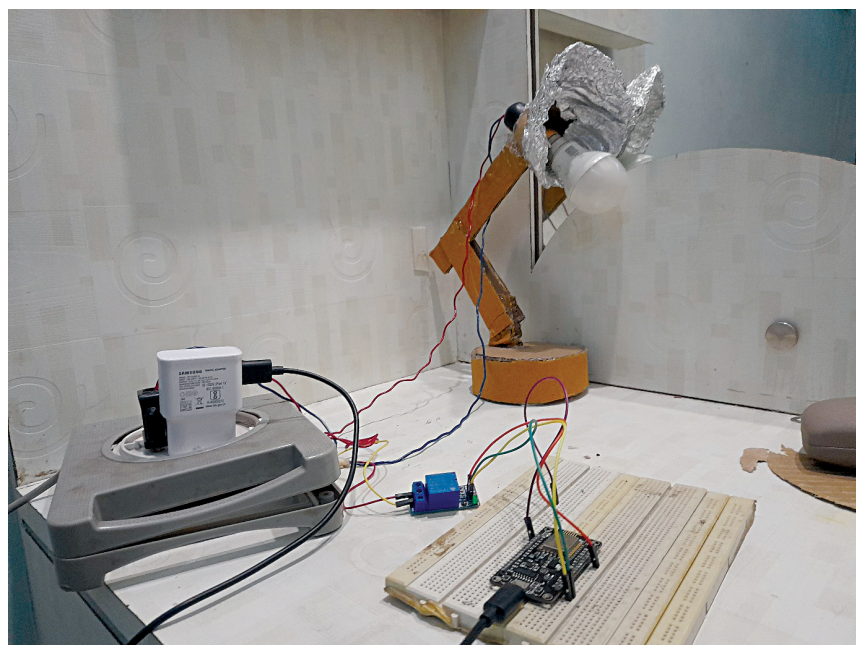


Fig. 5: Author's receiver circuit wired on breadboard

As soon as the state becomes high, the sensor data is processed by first NodeMCU that sends the data to smartphone and to the second NodeMCU in the receiver to turn on the lamp. When one gets up from the chair, the data from the transmitter is sent to the receiver to turn off the lamp.

The bed-time alert is set by the user in the smartphone through Blynk app. This set time is sent to Board2 to

turn off the lamp. Both the NodeMCU boards are connected to the Wi-Fi network or a mobile phone's hotspot.

Connection diagram of transmitter with IR sensor is shown in Fig. 1 and the pin-to-pin connections are shown in Table 1. Connection diagram of receiver with lamp is shown in Fig. 2 and pin connections in Table 2.

The AC lamp has line (L) and neutral (N) wires. Connect the neutral wire between normally-closed (NC)

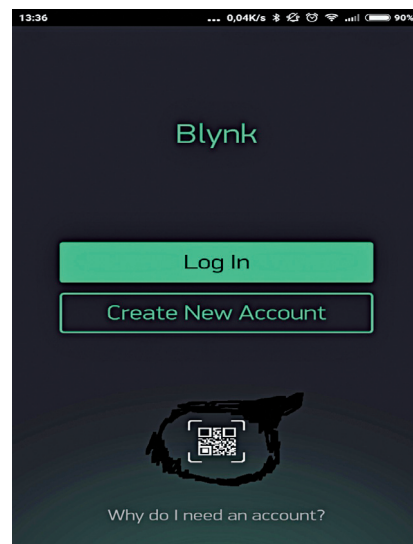


Fig. 6: Creating new account

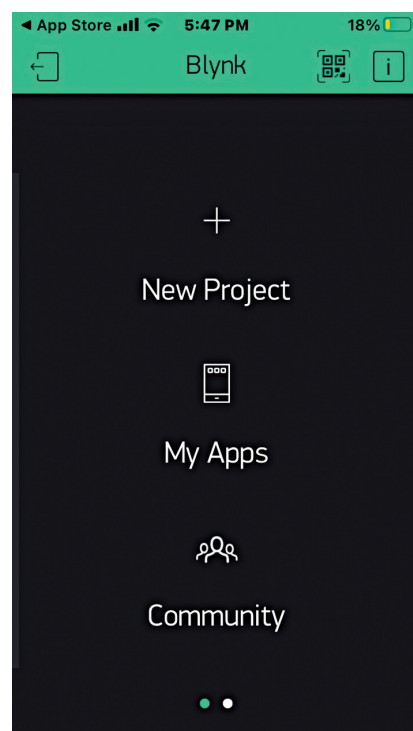


Fig. 7: Selecting New Project

pin and common pin of the relay module.

Software

Arduino IDE (integrated development environment) and Blynk app software are used in this project.

Arduino IDE. The Arduino IDE is the program used to write Arduino code. It comes in the form of a downloadable file on the Arduino website. The Arduino board is the